Jake Little

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PROFESSIONAL SUMMARY

M.S. in Aerospace and Mechanical Engineering with 2.5+ years of experience in design, simulation, and laboratory testing across wearable haptics, medical devices, marine vessels, and manufacturing. Successfully led six research projects while mentoring five undergraduate and high school students. Former Division I baseball athlete and recipient of multiple engineering scholarships.

SKILLS

Engineering: FEA (ANSYS, ABAQUS, FEBio), CFD (Fluent, OpenFOAM, Orca3D), Actuation systems and control, Electric motors and controllers, PIDs, Experimental design, Microcontrollers (Raspberry Pi, ESP32, Arduino)

Programming and Data: Python, C/C++, MATLAB, ROS2, LabVIEW, DEWESoft, Data acquisition and visualization

CAD and Prototyping: SolidWorks, Fusion 360, Rhino, 3D printing (FDM, SLA), Protoboards and PCB design, Welding

Soft Skills: Technical writing (thesis, journal papers, grants, documentation), Public speaking, Mentorship, Time management

EXPERIENCE

Mechanical Engineering Research Assistant — Aug 2023 – Present

CHROME Lab, Saint Louis University — *St. Louis, MO*

- Executed requirement-based design and testing cycles across nine devices in three user studies on haptic perception, achieving greater than 90% localization rates
- Reduced structural vibration by 71% through FEA modal analysis and validation using laser Doppler vibrometry
- Conducted mechanical durability and sound testing of additively manufactured enclosures to optimize design performance

Manufacturing Engineering Fellowship — May 2024 – May 2026

Lighthouse for the Blind of St. Louis — *Berkeley, MO*

- Designed and implemented seven accessible manufacturing technologies for blind and visually impaired employees
- Developed sensing and safety adaptations for chemical manufacturing lines spanning IR break beams, temperature dependent voltage dividers, serial data interfaces, and audible data reporting and alarms

Mechanical Engineering Intern — May 2023 – Aug 2023

Regal Boats — *Orlando, FL*

- Performed CFD analysis and validation on hull designs to assess running characteristics and excitation response
- Acquired and analyzed boat vibration and running data using DEWESOFT sensor and DAQ systems

PROJECTS

External Ventricular Drain (EVD) Clearance *Washington University in St. Louis* — Dec 2024 – May 2026

- Designed and developed a novel vibration-based catheter clearance system that doubled drain obstruction time by characterizing and optimizing 25 vibration propagation profiles for fluid clearance efficiency

Computational Fluid Dynamics for Microfluidic Flow Device — Jan 2024 – May 2024

- Leveraged simulation results for shear stress and fluid dynamics to guide device design and validate fluid conditions, ensuring physiologically relevant cell culture experiments

Infrared Photogate for Sickle Cell Confirmation — Oct 2021 – May 2023

- Engineered a low cost (<$25) automated data collection system and conducted a multiphase CFD study in OpenFOAM to evaluate alignment impact on droplet dynamics

EDUCATION

Master of Science (MS): Aerospace and Mechanical Engineering (with Distinction), Saint Louis University — May 2026

- Thesis: Design and Evaluation of Wearable Haptic Devices for Non-Visual Guided Breathing
- GPA: 4.00/4.00

Bachelor of Science (BS): Mechanical Engineering, Saint Louis University — May 2024

- GPA: 3.99/4.00, *summa cum laude*, Dean's List 2020-2024

AWARDS, HONORS & SCHOLARSHIPS

- Keynote Speaker (2024 SLU Scholarship Dinner), Treasurer of Tau Beta Pi Engineering Honor Society, co-inventor on a patent application, 1st Place Sigma Xi Poster Presentation, 2nd Place Rice 360 Global Health Challenge
- Recipient of seven scholarships, including five engineering and two university/athletic scholarships